KUANBUTTS.COM
KUANBUTTS@GMAIL
CURRICULUM VITAE

# KUAN BUTTS

geomatics, spatial computation, network science, OSS, data science/engineering

## EXPERIENCE

### **Mapbox** — *Staff Software Engineer*

NOV 2019 - PRESENT, SAN FRANCISCO, CA

*SENIOR SWE from NOV 2019 - APR 2025*

- Processing and classification of live global stream of multimodal location data
- Calculation of real-time road speed profiles, feeding ETA and route selection
- Developed distributed inference pipeline for real-time global road segment speed estimates, handling >5B observations/hour
- Migrated live speeds to decoupled, high-accuracy probe-matching system, improving ETA precision by +5.9% and recall by +10.8%. Optimized modality classifier, achieved speed accuracy F1 gains of 2-11% (region dependent) via revamped matching system; yielded up to ~60% increased coverage on surface roads
- Streamlined pipelines for modality classification model training on raw probe data. Developed ML applications on vehicle telemetry to flag routing anomalies.
- Delivered a global, daily-updated activity indices product at road segment and block-level resolution, enabling high-granularity mobility insights.

### **UrbanFootprint** — *Data Science & Engineering*

JAN 2017 - SEP 2019, BERKELEY, CA

- Optimization of high resolution spatial models with prolonged runtimes
- Development of network graph models on urban travel network data

### **Code for America** — *Software Engineer, Fellow*

JAN 2016 - DEC 2016, SAN FRANCISCO, CA

- Text-based communications software development (ClientComm)
- Generated large text communications dataset for client success rate research
- Application deployed in 3+ US jurisdictions, serving thousands of end users

### **Microsoft** — *Civic Technology Fellow*

AUG 2015 2016 - DEC 2015, NEW YORK CITY, NY

- Ingestion of live NYC MTA bus data to monitor real-time schedule adherence

### **MIT + MIT Media Lab** — *Graduate Researcher, Software Developer*

AUG 2012 - AUG 2015, CAMBRIDGE, MA

- Contributed to development of app to geotag rider sentiment along transit lines

## EDUCATION

### **MIT** — *MCP, Transportation Systems Planning*

AUG 2012 - SEP 2014, CAMBRIDGE, MA

Thesis: *Iterative Methods in Adapting Mobile Technologies for Data Acquisition, A Case Study in St. Louis, Missouri*